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ID: 15691767 **Date of birth**: 23/07/1998

Place of birth: Vila Franca de Xira, Portugal **Nationality**: Portuguese

TECHNOLOGICAL STACKS

[20/07/2024 – Current] **AI**

Projects:

PoC_Plug&Play - Use RAG to answer complex questions on multiple web, local and MCP sources for specific use-cases;

PoC_StopSales - Start or stop hotel chain sales from emails;

PoC_PriceComparison - Compare online prices with AI Agents web searches;

PoC_RAG - Question/Answer to email threads with attachments;

PoC_4 - AI data extraction from Excel and CSV files;

PoC_3 - AI categorizer and modifier of Excel and CSV files;

PoC_UiPath_CM_TM - PoC of UiPath Communications Mining + Task Mining for AI automation of emails and repetitive tasks;

PoC_Secil - PoC Secil (with Azure AI Computer Vision) for detecting PPE misuse;

Knowledge:

- OpenAI;
- Fine-tuning OpenAI models;
- LangChain;
- LangGraph;
- LangFlow;
- LangSmith;
- Python;
- RAG (web, email, docx, pdf, txt, excel, csv, images);
- VectorDB & Embeddings;
- HybridSearch (Ensemble Retrievers, BM25, ANN);
- ElasticSearch;
- Re-rankers;
- MCP (Fully customized MCPs, SQL Server, MySQL, MongoDB);
- Tavily, Brave, Perplexity, BrowserAct, Crawl4Ai, FireCrawl, Playwright;
- N8N;
- Local fine-tuning (WSL, PyTorch, CUDA, Unsloth, llama.cpp, Ollama, LMStudio, Quantization Methods, LoRa and QLoRa, train with validation, tests with different models, datasets and quantizations);
- Q/A Agents custom performance metrics;
- HuggingFace;
- Azure OpenAI;

- Azure AI Computer Vision;
- UiPath Communications Mining + Task Mining;

[04/07/2021 – 30/12/2024] **Flutter & Dart**

Projects:

Bluenergy - APP for managing and analyzing electricity, gas, fuel and water consumption;

Xpend - APP for managing business costs;

FitAPP - APP for monitoring health, sleep and physical activities;

Omnio - APP for controlling and managing employee productivity;

Planet Agora - WhatsApp/Discord-style chat and video calling APP with moderation for minors and gamification;

BoilerplateFlutter - APP that serves as base for all Flutter projects;

Gudnest - Bird management app;

GeoAds - Georeferenced advertising APP;

Dinder - Cooking Recipes APP;

TheBigHand - APP for fundraising and aid;

MyAPIR - APP for renal failure management;

MBWhere - APP that allows you to locate and check malfunctions in ATMs;

InspirePeople - News and events APP for the InspireIT company;

State Managements:

- MobX;
- Provider;
- ChangeNotifier;

Knowledge:

- SOLID & Clean Architecture;
- MVC and Modular Architecture;
- Null-Safety;
- Deploy and manage on Google and Apple stores;
- Flutter Mobile & Web;
- Android & iOS Flavors/Environments management;
- Retry Policy;
- Unit, Widget and Integration Tests;

Some implemented Features:

- **about** - Company and project information, APP version, redirection to the email APP, WebViewer inside the APP, and redirection of the destination address to the user's preferred mapping APP (GoogleMaps, Waze, etc);
- **app_lifecycle** - Controls background or foreground async functions;
- **assets** - Images (PNG and SVG) and Fonts;
- **authentication** - Handles registration, login, password recovery, forms, providers, http requests, biometric login, OTP, etc;
- **biometrics_unlocks** - Biometric authentication;
- **connectycube** - Controls the chat and videocall service;

- **device_info** - Device Specifications;
- **encrypted_secure_storage** - Encrypted secure storage;
- **file_picker** - Picker of files from the gallery, camera or device memory;
- **firebase_crashlytics** - Firebase Crashlytics;
- **fl_chart** - Various types of customizable graphics;
- **geolocation** - Geolocation of the device;
- **global_context** - Way to get the BuildContext anywhere in the code;
- **http** - Most commonly used HTTP methods and requests;
- **introduction** - Module that contains the introduction of the APP when it is the first run;
- **launch_icon** - Customized APP Icon;
- **maps** - Use of Maps with GeoLocation;
- **mixpanel** - MixPanel APP Analytics;
- **multilanguage** - Multilanguage picker saved in local storage;
- **settings** - Settings screen that has the general, user account settings and some extra features;
- **shared_preferences** - Used to store the APP's preference settings, in local storage;
- **splash** - Splash screen (native and non-native) that appears during APP startup, in order to load the relevant data for its proper functioning;
- **themes** - App Themes (adaptive according to the OS theme);
- **uni_links** - Handles UniLinks (DeepLinks) to externally open the app in the desired location;
- **video_player** - Video Player;
- **webview** - WebViewer inside the APP that allows you to access an external site;
- **youtube_player** - Youtube video player in APP;

[07/2021 – 04/2022] **C# / .NET Core**

Projects:

MBWhere - The ATMs Waze;

Backend knowledge:

- Onion and Multi-Layered Architectures;
- Use of REST, JSON and HTTP;
- Creating Web APIs;
- Routing, filters, asynchronous methods;
- Authentication, authorization, and communication with JWT (JSON Web Token);
- Migrations;
- AutoMapper;

[07/2021 – 04/2022] **Data Bases**

DBs [Projects]:

SQL Server [MBWhere];

Firebase [InspirePeople];

MongoDB [BluSensor];

Knowledge:

- Queries;
- Views;
- Functions;

- Cursors;
- Procedures;
- Triggers;

[08/2019 – 06/2022] **C / C++**

Projects:

SmartHome - Control of the house through an APP:

- A smart electrical panel was developed that measured the power of the components it controlled, these being the lights and the floor heaters.
- An APP was developed in "MIT APP Inventor" that contained the house plan and its controllable components.

BluSensor - Consumed power measurement sensor:

- measured the consumed power through light beams collected from the meter of the contracted service;
- automatically sent all collected and calculated values to the API and e-mail;
- this device contains a battery and can be charged;

BluSensor Calha DIN - DIN rail power consumption measurement sensor:

- measured the consumed power directly from the electric current;
- this device is placed on the DIN rail of the electrical panel to facilitate immediate access to the electrical current to be measured;

DrVidaPocket - Small medical device with the purpose of performing PCR exams:

- after the sample is taken and placed in the device, the device can identify if it is a positive or negative test;
- automatically sends the result to the APP, the user's e-mail and the National Health System;

Some of the implemented / used features:

- Restructuring the project architecture into modules and layers;
- Programming in C / C++ and Assembly;
- Microcontrollers used: Arduino and ESP32;
- Use of Wi-Fi modules;
- Use of Bluetooth and BLE modules;
- Mail server for microcontrollers;
- Collecting data from sensors;
- Use of digital and analog actuators;
- Configuring and using MongoDB;
- Calls to APIs;
- Elaboration of Printed Circuit Boards (PCBs);
- Elaboration of electrical circuits;

[08/2019 – Current] **Software, IDEs and Extensions**

Versioning Managers:

- Bitbucket;
- GitHub;
- GitLab;
- GitKraken;
- SourceTree;

Task Managers:

- ClickUP;
- Jira;
- Trello;
- YouTrack;

Software, IDEs and Extensions:

- Android Studio;
- Arduino IDE;
- Claap;
- Docker;
- Draw.io;
- Figma;
- Fritzing;
- GitHub Copilot (AI pair programmer);
- Microsoft SQL Server Management Studio;
- OpenVPN;
- Oracle VM VirtualBox;
- PlatformIO;
- Postman;
- Visual Studio;
- Visual Studio Code;
- XCode;

[08/2019 – Current] **Methodologies**

Agile:

- Kanban;
- Scrum;

Others:

- Waterfall;

WORK EXPERIENCE

InspireIT

City: Caparica | **Country:** Portugal

[04/07/2021 – Current] **Flutter Fullstack Software Developer**

AI development with Python and low-code solutions.

Mobile applications development with Flutter.

Server-level development with .NET Core, C#, WebAPI, SQL Server and Docker.

Firmware level development for IoT with C/C++.

Development of layered architectures.

Development of support tasks.

Understanding the workflow associated with development methodologies (Kanban and Scrum) using Jira and ClickUp.

Version control with Git and Bitbucket.

Technical Documentation.

Bluenergy

City: Lisboa | **Country:** Portugal

[08/2019 – 31/08/2021] **Developer of programmable electronic devices**

Independent part-time work while I was finishing my degree.
Development of consumption monitoring systems.
Dimensioning of electrical and electronic circuits.
PCB development.
Welding of very small dimension components (SMD).
Programming in C/C++ (Arduino).

VisionVolt

City: Caparica | **Country:** Portugal

[01/2019 – 06/2019] **Electronics Technician, Embedded Systems Programmer and 3D Parts Designer**

Dimensioning of electrical and electronic circuits.
Investigation of circuits and solutions, at electronics and programming levels.
Development of complex projects for end customers.
PCB development.
Welding of very small dimension components (SMD).
Programming in C/C++ (Arduino) and Python (RaspberryPi).
Development of home automation platforms.

CTDI

City: Lisbon | **Country:** Portugal

[02/2016 – 06/2016] **Mobile phone repair technician and Quality Control**

Internship.
Quality Control of mobile phones and routers.
Repair of various brands of mobile phones.

EGEA

City: Cosenza | **Country:** Italy

[24/06/2015 – 24/07/2015] **3D Modeler**

International internship in Italy sponsored by Erasmus+.
Development of 3D modeling of buildings, mostly using Cinema4D.

EDUCATION AND TRAINING

[18/08/2022 – 18/08/2022] **Course "Dart - Advanced Course"**

Udemy

- Working with operating system variables (OS variables, running processes, communicating with processes);
- Async Programming (timers, callbacks, futures, await);
- Compression (gzip and zlib compression, zip files);
- Encryption (hashes, deriving keys, secure random numbers, stream ciphers, block ciphers);

- Socket programming (TCP server and client, HTTP, UDP);
- Database programming (database design, selecting rows, queries, transactions);

[18/08/2022 – 18/08/2022]

Course "Dart - Intermediate Course"

Udemy

- Imports;
- Classes;
- Scope (public, private, getters, setters, static);
- Polymorphism (inheritance, mixins, interfaces, abstraction);
- Generics;
- File system (sync, async, temp directories, listing and accessing directories and files);

[25/11/2021 – 02/12/2021]

Course "Null-Safety in Dart/Flutter 2.0 - Learn in Detail - 2021"

Udemy

- Understand what is Null-Safety and its Principles;
- Apply Null-Safety concepts in Dart/Flutter;
- Make Dart/Flutter code safer and more reliable with Null-Safety;
- Automatically migrate Dart/Flutter code to Null-Safety;
- Advanced Null-Safety;
- Course completion.

[05/05/2021 – 22/05/2021]

Course "Learn Flutter and Develop Apps for Android and IOS 2021"

Udemy

- Flutter Fundamentals;
- Dart - Fundamentals;
- Dart - Control Structures;
- Dart - Functions;
- Dart - Object Orientation;
- Basics of Flutter;
- Debugging & Device Differences;
- Building Real App - Personal Expenses;
- Responsive & Adaptive Interfaces;
- Flutter & Widgets Internally;
- Navigation & Multiple Screens;
- State Management;
- Working with Forms;
- Sending HTTP Requests;
- Adding Authentication;
- Adding Animation;
- Using Native Resources (Camera, Maps, BD, ...);
- Chat application - Firebase, Image Upload and Push Notifications;
- Running Native Code on Android and iOS;
- Calculator project;
- Publishing the Application in App Stores;

- Minesweeper Project - Flutter Course;
- Course Completion;

[14/04/2021 – 21/04/2021] **Programming course with App Inventor**
CDI Portugal and RECODE

[13/04/2021 – 13/04/2021] **Technologies for the Future course**
CDI Portugal and RECODE

[22/03/2021 – 05/07/2021] **Co-Care Course - Development in Co-creation of Health Apps**
Polytechnic Institute of Setubal (EST e ESS) with the collaboration of Vic University

City: Setúbal | **Country:** Portugal | | **Type of credits:** ECTS | **Number of credits:** 4

The student will develop a set of activities that will allow him to learn about:

- (i) the use of technology in the health context;
- (ii) the fundamental aspects to consider in the development of ICT (Information and Communication Technologies) in health areas;
- (iii) how to develop a project using the co-creation methodology.

The student will develop as a practical project a prototype of a new software application for health following the co-creation model.

The prototype to be developed will be a support app for Alzheimer's caregivers and the development team will integrate students from the areas of technology and health and caregivers of Alzheimer's patients.

General notes:

Lasting 108 hours in 15 weeks and equivalent to 4ECTS, the course will be in b-learning mode and will have students from the technology and healthcare areas.

[09/2019 – 10/2021] **Degree in Computer Engineering, Software branch**
Polytechnic Institute of Setubal

City: Setúbal | **Country:** Portugal | | **Final grade:** 15 | **Level in EQF:** EQF level 6 | **Type of credits:** ECTS | **Number of credits:** 180

General:

- Mathematics I;
- Numerical analysis;
- Discrete Mathematics;
- Statistical methods;
- Economy;
- Introduction to Business Management;
- Organizational Behavior and Team Management;
- Innovation and Entrepreneurship;

Specific:

- Computational Architecture and Microprocessors;
- Introduction to Object Oriented Programming (Java);
- Computer Networks (Cisco);
- Algorithms and Abstract Data Types (C);

Databases (My SQL);
 Introduction to Web Programming (HTML and CSS);
 Object Oriented Programming (Java and JavaFX);
 Advanced Programming (Java and JavaFX);
 Web programming (Javascript, Node.js, Bootstrap, Ajax, Express, REST, MongoDB);
 Operating Systems (Xubuntu, C);
 Database Complements (SQL Server);
 Mobile Computing (AndroidStudio - Java, React Native);
 Parallel and Distributed Computing (Python);
 Person-Machine Interaction (AdobeXD, Figma);
 Video Game Development (Unity - C#);
 Applied Software Engineering (Jira, Confluence, Bitbucket);
 Visual Programming (C#, ASP.NET, MVC, MS SQL, Angular, HTML and CSS);
 Computer Project Management;
 Artificial Intelligence (Lisp);
 Security and Information Technology (SEEDUbuntu - C);
 Portfolio in Computer Engineering.

[01/2017 – 03/2017] **Digital Marketing Certificate**

Google

City: Setúbal | **Country:** Portugal |

[09/2016 – 07/2019] **Professional Senior Technician in Electronic and Computer Systems**

Polytechnic Institute of Setubal

City: Setúbal | **Country:** Portugal | | **Final grade:** 16 | **Level in EQF:** EQF level 5

General:

Elements of Mathematics 1;

Elements of Mathematics 2;

Technical English;

Business Management and Organizational Behavior.

Professional:

Signal Acquisition and Processing;

Computer Systems Architecture;

Analog Electronics;

Electrotechnics;

Introduction to Programming (C/C++);

Introduction to Telecommunications;

Microcontrollers (Arduino);

Object Oriented Programming (Java and Java FX);

Electronic Systems Project;

Computer network;

Sensor Networks;

INETE

City: Lisboa | **Country:** Portugal | | **Final grade:** 17 | **Level in EQF:** EQF level 4

Digital and Analog Electronics;

Microcontrollers programming (C/C++ language, ATMEL family microcontrollers, Arduino and Raspberry PI);

Automaton Programming: Zelio manufacturer Telemecanique, S7-1200 manufacturer SIEMENS in LADDER and Grafset language;

C/C++ programming;

Python programming;

Programming and Development of APPs for Android OS;

Industrial and Mobile Robotics;

Home Automation (x.10 and EIB networks);

Pneumatics and Hydraulics;

Technical drawing based on AutoCAD;

Analysis and interface of electronic circuits in a virtual environment (Multisim and Proteus);

Data acquisition systems (Labview);

Electronic Through Hole Welding;

Data networks (Automation and home automation networks);

Design of printed circuit boards.

LANGUAGE SKILLS

Mother tongue(s): Portuguese

Other language(s):

English

LISTENING C1 READING C1 WRITING C1

SPOKEN PRODUCTION C1 SPOKEN INTERACTION C1

Italian

LISTENING B1 READING B1 WRITING A1

SPOKEN PRODUCTION A2 SPOKEN INTERACTION A2

Spanish

LISTENING B1 READING B1 WRITING A1

SPOKEN PRODUCTION A2 SPOKEN INTERACTION A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

PUBLICATIONS

[2023]

[Integrate UniLinks with Flutter \(Android AppLinks + iOS UniversalLinks\)](#)

Step-by-step guide of how to integrate UniLinks using Flutter Mobile and Flutter Web.

Article in english and available in: <https://medium.com/@pedrostick3/integrate-unilinks-with-flutter-android-applinks-ios-universallinks-c9a1542d6625> .

DRIVING LICENCE

Cars: B1
Cars: B

RECOMMENDATIONS

During the TEAC Course (Electronics, Automation and Command Technician), I obtained, in the 3 consecutive years, in every quarter, Certificates of:

- Attendance (Only receives those who have 0 Hours of Excused and Unexcused Absences);
- Merit (Only those who have an average above 15 points are awarded and those who have not had unjustified absences).

I finished a degree in Computer Engineering which lasts for 3 years in 2 years, while working for the company Bluenergy and while taking several courses (Course Technologies for the Future, Course Programming with App Inventor, Course "Learn Flutter and Develop Apps for Android and IOS 2021" and Co-Care Course - Development in Co-creation of Health Apps).

INTERNSHIPS

Internships

At LEI (Degree in Computer Engineering [Level 6]) I did an internship at InspireIT, which is a software development company that creates digital products that make life easier for the user through different approaches, in order to offer more quality and better service, from dedicated and personalized way to its customers.

At the CTeSP-SEC (Professional Superior Technical Course in Electronic and Computer Systems [Level 5]), I did an internship at the company VisionVolt, which is a company that provides services in the scope of development of electronic systems and programming.

In my 12th year, that is, 3rd year at CP-TEAC (Professional Course for Electronics, Automation and Command Technician [Level 4]), I did an internship at CTDI which is a mobile phones and router repair company doing Quality Control and Repair.

In my 11th year, that is, 2nd year of CP-TEAC (Professional Course for Electronics, Automation and Command Technician [Level 4]), I interned at a civil construction company, located in Italy and called EGEA, for a month, in the Architecture department, sponsored by the Erasmus+ program, where I worked with AutoCAD, ArchiCAD and Cinema 4D, getting a good performance.

CONTESTS AND DISCLOSURE

Contests and Disclosure

I've been in several robotics, electronics and programming contests, such as "Faqtos", where I got 1st place, "RoboParty", where I got 2nd place, "aptipro", among other contests.

I've been in "Futurália" and "MakerFair" several times with the aim of publicizing the projects of the school where I was (INETE).

My Proof of Professional Aptitude (PAP) was published in the magazine "Robótica Nº 105" which finds facts in the field of technologies.

I have always developed my own personal projects, as well as school ones, having done more than 15 projects to date.

WORK RELATED SKILLS

Work related skills

I am committed, very persistent and I have a good mastery in programming (Dart, Flutter, C/C++, C#, Java, Java FX, SQL, Arduino and Ladder) and 3D Modeling.

I also know how to make complex electrical circuits.

OTHER COMPETENCES

Other competences

"When I grow up I want to be an inventor scientist and I want to improve people's lives!" - Typical phrase from when I was a minor and I still have the same motto. Always studying and making new and improved projects.

I like to help and find solutions to all kinds of problems.

Persistence, determination and perfectionism are my strengths.

COMMUNICATION AND INTERPERSONAL SKILLS

Communication and interpersonal skills

I practiced Roller Hockey for 8 years.

I was at ESTuna (Tuna of the Superior School of Technology of Setúbal).

I worked for a Recommendation Marketing company.

ORGANISATIONAL SKILLS

Organisational skills

I am organized and have a good coordination ability when it comes to group activities, also a little bit due to having practiced roller hockey in several clubs such as "Sporting", "Sport Alenquer e Benfica" and "Alverca".

I like to help those who have difficulties.